

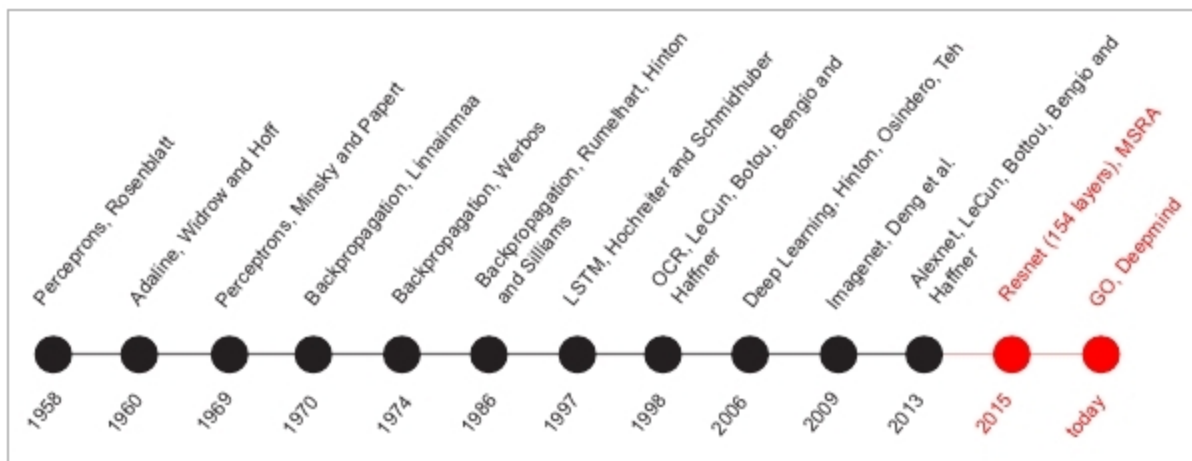


Figure 6: Evolution of artificial neural networks and deep learning

learning. These include Google, Amazon, Microsoft, Facebook, Yahoo, Samsung, Xerox, NVIDIA and Baidu, to name a few. The growth of ML can be assessed from the fact that several courses on it are being offered by reputed universities on MOOC platforms and job offerings in this field are on the rise. The technologies being developed, including the hardware (such as NVIDIA GPU cards), have also arrived in the market.

### Deep learning: A brief overview

Deep learning is a recent development in the field of ML. It has enabled researchers to design algorithms that can run in parallel on GPUs and achieve quick results on huge amounts of training data. Currently, the world is populated with data everywhere—in different formats such as text, audio, image, and video—which includes spatial data, e-commerce data, surveillance data, social network data and so on. Handling such large volumes of data was not possible some years ago, since the systems in those days were 'data rich, but information poor'. GPUs have solved this

issue, and mammoth sized computations can now be carried out in a limited amount of time. Li Deng mentions [see Reference 2 at the end of the article] that deep learning has become enormously popular nowadays due to the availability of great processing power (GPUs), low cost hardware, and advances in the ML and signal processing domains. All the giant companies worldwide have moved towards deep learning based solutions. And many innovative, small startups (such as Nuance, Clarifai, etc) are following suit.

### Applications of deep learning

Listed below are some of the interesting applications of deep learning:

- Skype Language Translator can translate one language into another, in real-time.
- Google Deep Mind has recently defeated the human champion of the Go game.
- Facebook face recognition can recognise the same individual in different photos with almost 97 per cent accuracy. END 🐧

### References

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- [3] E Gavves; 'Introduction to neural networks and deep learning', a UVA deep learning course.
- [4] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton; 'Deep learning', *Nature Magazine*, May 2015

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